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RESOLUTION MEPC.214(63)
Adopted on 2 March 2012
2012 GUIDELINES ON SURVEY AND CERTIFICATION OF THE
ENERGY EFFICIENCY DESIGN INDEX (EEDI)RESOLUTION MEPC.214(63)
Adopted on 2 March 2012
2012 GUIDELINES ON SURVEY AND CERTIFICATION OF THE
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SOURCE: MEPC.215(63).pdf
RESOLUTION MEPC.215(63)
Adopted on 2 March 2012
GUIDELINES FOR CALCULATION OF REFERENCE LINES FOR USE WITH THE
ENERGY EFFICIENCY DESIGN INDEX (EEDI)MEPC 63/23/Add.1
Annex 11, page 1

I:\MEPC\63\23-Add-1.doc
ANNEX 11

RESOLUTION MEPC.215(63)
Adopted on 2 March 2012

GUIDELINES FOR CALCULATION OF REFERENCE LINES FOR USE WITH THE
ENERGY EFFICIENCY DESIGN INDEX (EEDI)

THE MARINE ENVIRONMENT PROTECTION COMMITTEE,

RECALLING Article 38(a) of the Convention on the International Maritime Organization concerning the functions of the Marine Environment Protection Committee (the Committee) conferred upon it by international conventions for the prevention and control of marine pollution,

RECALLING ALSO that, at its sixty-second session, the Committee adopted, by resolution MEPC.203(62), amendments to the Annex of the Protocol of 1997 to amend the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto (inclusion of regulations on energy efficiency for ships in MARPOL Annex VI),

NOTING that regulation 21 (Required EEDI) of MARPOL Annex VI, as amended, requires

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of 1978 relating thereto (inclusion of regulations on energy efficiency for ships in MARPOL Annex VI),

NOTING that regulation 21 (Required EEDI) of MARPOL Annex VI, as amended, requires reference lines to be established for each ship type to which regulation 21 is applicable,

HAVING CONSIDERED, at its sixty-third session, the draft Guidelines for calculation of reference lines for use with the Energy Efficiency Design Index (EEDI),

ADOPTS the Guidelines for calculation of reference lines for use with the Energy Efficiency Design Index (EEDI), as set out at annex to the present resolution.

RESOLUTION MEPC.215(63)
Adopted on 2 March 2012
GUIDELINES FOR CALCULATION OF REFERENCE LINES FOR USE WITH THE ENERGY EFFICIENCY DESIGN INDEX (EEDI)MEPC 63/23/Add.1
Annex 11, page 2

I:\MEPC\63\23-Add-1.doc
ANNEX

GUIDELINES FOR CALCULATION OF REFERENCE LINES FOR USE WITH THE ENERGY EFFICIENCY DESIGN INDEX (EEDI)

1 The reference lines are established for each ship type to which regulation 21 (Required EEDI) of MARPOL Annex VI is applicable. The purpose of the EEDI is to provide

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ENERGY EFFICIENCY DESIGN INDEX (EEDI)

1 The reference lines are established for each ship type to which regulation 21 (Required EEDI) of MARPOL Annex VI is applicable. The purpose of the EEDI is to provide a fair basis for comparison, to stimulate the development of more efficient ships in general and to establish the minimum efficiency of new ships depending on ship type and size. Hence, the reference lines for each ship type is calculated in a transparent and robust manner.

2 Ship types are defined in regulation 2 of MARPOL Annex VI. The reference line for each ship type is used for the determination of the required EEDI as defined in regulation 21 of MARPOL Annex VI.

3 These guidelines apply to the following ships types: bulk carrier, gas carrier, tanker, containership, general cargo ship, refrigerated cargo carrier and combination carrier. It is noted that a method of calculating reference lines has not been established for ro-ro cargo ships, ro-ro cargo ships (vehicle), ro- ro-passenger ships and passenger ships, as well as for ships with diesel-electric propulsion, turbine propulsion and hybrid propulsion.

Definition of a reference line

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ships, ro-ro cargo ships (vehicle), ro- ro-passenger ships and passenger ships, as well as for ships with diesel-electric propulsion, turbine propulsion and hybrid propulsion.

Definition of a reference line

4 A reference line is defined as a curve representing an average index value fitted on a set of individual index values for a defined group of ships.

5 One reference line is developed for each ship type to which regulation 21 of MARPOL Annex VI is applicable, ensuring that only data from comparable ships are included in the calculation of each reference line.

6 The reference line value is formulated as Reference line value = a (100% deadweight) -c where "a" and "c" are parameters determined from the regression curve fit.

7 Input data for the calculation of the reference lines is filtered through a process where data deviating more than two standard deviations from the regression line are discarded. The regression is then applied again to generate a corrected reference line. For the purpose of documentation, discarded data is listed with the ships IMO number.

Data sources

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discarded. The regression is then applied again to generate a corrected reference line. For the purpose of documentation, discarded data is listed with the ships IMO number.

Data sources

8 IHS Fairplay (IHSF) database is selected as the standard database delivering the primary input data for the reference line calculation. For the purpose of the EEDI reference line calculations, a defined version of the database is archived as agreed between the Secretariat and IHSF.

9 For the purpose of calculating the reference lines, data relating to existing ships of 400 GT and above from the IHS F database delivered in the period from 1 January 1999 to 1 January 2009 are used.

RESOLUTION MEPC.215(63)
Adopted on 2 March 2012
GUIDELINES FOR CALCULATION OF REFERENCE LINES FOR USE WITH THE ENERGY EFFICIENCY DESIGN INDEX (EEDI)MEPC 63/23/Add.1
Annex 11, page 3

I:\MEPC\63\23-Add-1.doc
10 T he following data from the IHSF database on ships with conventional propulsion systems is used when calculating the reference lines:

.1 data on the ships' capacity is used as Capacity for each ship type as defined in MEPC.XXX(63);

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SOURCE: MEPC.215(63).pdf
10 T he following data from the IHSF database on ships with conventional propulsion systems is used when calculating the reference lines:

.1 data on the ships' capacity is used as Capacity for each ship type as defined in MEPC.XXX(63);

.2 data on the ships' service speed is used as reference speed Vref; and

.3 data on the ships' total installed main power is used as MCRME(i).

11 For some ships, some data entries may be blank or contain a zero (0) in the database. Datasets with blank power, capacity and/or speed data should be removed from the reference line calculations. For the purpose of later references, the omitted ships should be listed with their IMO number.

12 To ensure a uniform interpretation, the association of ship types defined in regulation 2 of MARPOL Annex VI, with the ship types given by the IHSF database and defined by the so-called Stat codes, is shown in the appendix to this guideline. Table 1 in the appendix lists the ship types from IHSF used for the calculation of reference lines. Table 2 lists the IHSF ship types not used when calculating the reference lines.